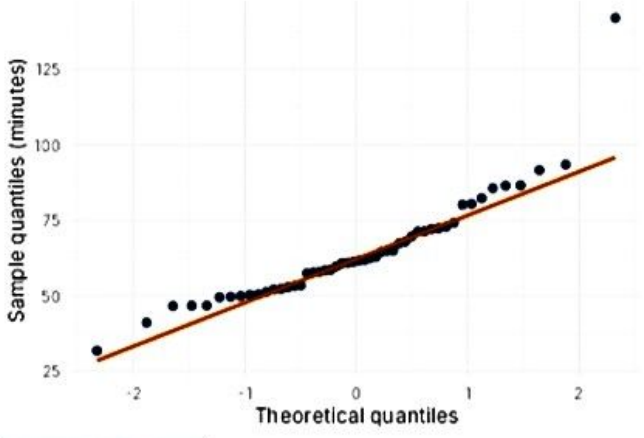
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

R.No: 15 Reg. No: 1925260144 Name: G Ganga Chandrika

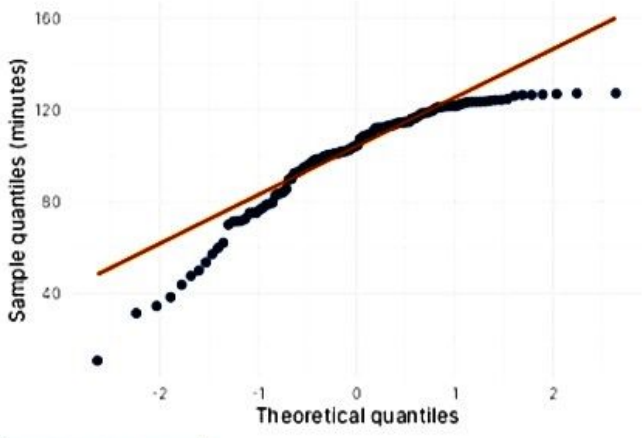
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: daily study time (minutes), n = 50



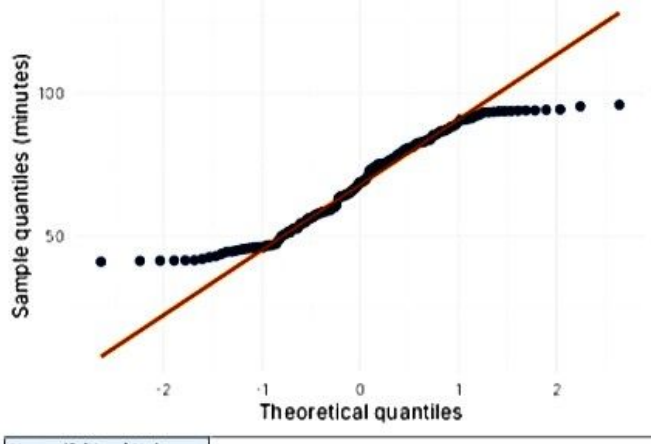
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: daily study time (minutes), n = 120



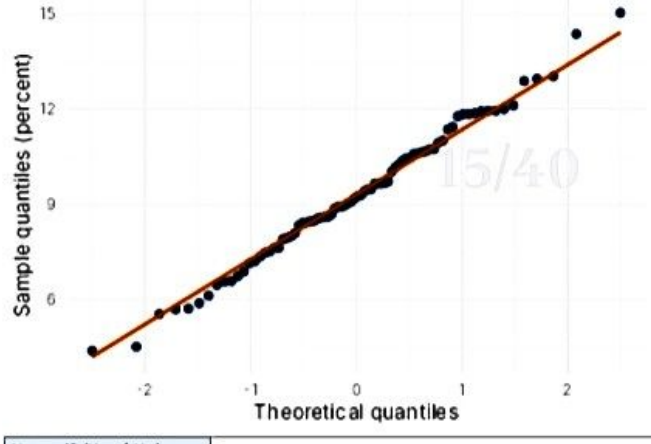
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: daily study time (minutes), n = 120



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: battery drain per hour (percent), n = 80



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Normal Q-Q plot Analysis-

Chart - 1

Daily study time (min) n=50

1. Normal? No
2. Shape : Right - Skewed
3. Evidence in plot :

Most points follow the line, but the upper tail has points above the line, including one extreme high value.

Chart - 2

- Normal ? No
- Shape (from the list) : Left - Skewed
- Evidence in the plot : The points curve away from the reference line, with a longer lower tail & the upper points flattering below the line.

Chart - 4

- Normal ? NO
- Shape (from the list) : Light - tailed
- Evidence in the plot : Most points lie close to the straight reference line, with only small deviation at the tails.

Chart - 4

- Normal ? Yes
- Shape (from the list) : Normal
- Evidence in the plot : Most points to lie close the straight reference line, with only small deviations at the tails.

Part - B

B_1 - Chart - 1 shows right - skewness (positive skewness) meaning the data is not normally distributed. A log transformation can be used to reduce the skewness before applying mean - & - standard deviation methods.

B_2. The Blank are;

1. stat_qq()
2. stat_qq_line()